
At Most One – AMO, At Least One – ALO Encoding

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Outline

1. Binomial encoding
2. Binary encoding
3. Sequential encounter encoding
4. Commander encoding
5. Product encoding

AMO, ALO Encoding

- Applications: i.e., Sudoku, Hitori, Numberlink,...

- x_{ijk} iff k at (i,j) where $1 \leq i, j, k \leq 9$

- Each box contains precisely one number.

$$(x_{ij1} \vee x_{ij2} \vee \dots \vee x_{ij9}) \wedge (x_{ij1} \supset \neg x_{ij2}) \text{ (i.e., } \neg x_{ij1} \vee \neg x_{ij2}), \dots$$

- Precisely once in each row (fix i):

$$(x_{i1k} \vee x_{i2k} \vee \dots \vee x_{i9k}) \wedge (x_{i1k} \supset \neg x_{i2k}) \text{ (i.e., } \neg x_{i1k} \vee \neg x_{i2k}), \dots$$

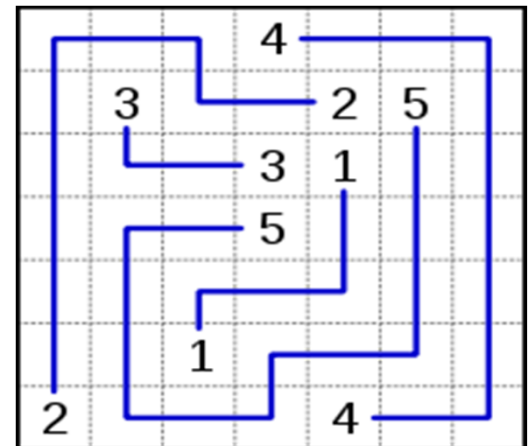
- Precisely once in each column (fix j):

$$(x_{1jk} \vee x_{2jk} \vee \dots \vee x_{9jk}) \wedge (x_{1jk} \supset \neg x_{2jk}) \text{ (i.e., } \neg x_{1jk} \vee \neg x_{2jk}), \dots$$

- Precisely once in each sub-3x3 matrix.

$$(x_{11k} \vee x_{12k} \vee \dots \vee x_{33k}) \wedge (x_{11k} \supset \neg x_{12k}) \text{ (i.e., } \neg x_{11k} \vee \neg x_{12k}), \dots$$

5								
	4					2	9	
		8	2			4		5
		4		9			5	3
			1		2			
				4		7		
	2	5			6	1		
	3		5				7	
		7	8					6



AMO – At Most One Truth; ALO – At Least One Truth

AMO – Binomial encoding

- Variables: the same $x_0, \dots, x_n$
→ no new variables
- Clauses: for $0 \leq i < j < n$, $\neg x_i \vee \neg x_j$ ($x_i \rightarrow \neg x_j$)
- Requires: $O(n^2)$ clauses

ALO: $x_0 \vee x_1 \vee \dots \vee x_n$

Binary encoding

- Idea:
 - Variables: X_i and new k variables $Y_1, Y_2, \dots, Y_k$ ($k = \lceil \log_2 n \rceil$)
 - X_i corresponding a binary string
 - Clauses: $X_i \leftrightarrow \varphi(i,1) \wedge \varphi(i,2) \wedge \dots \wedge \varphi(i,k)$ where
 - $\varphi(i,j)$ denotes for Y_j if *j-th* digit in binary string is 1
 - $\varphi(i,j)$ denotes for $\neg Y_j$ otherwise
 - Requires: $O(\log_2 n)$ new variables, $O(n \cdot \log_2 n)$ clauses

Binary encoding (1)

- For $n = 4$ (X_1, X_2, X_3, X_4)

X_1 corresponding “00”

$$X_1 \leftrightarrow \neg Y_1 \wedge \neg Y_2$$

X_2 corresponding “01”

$$X_2 \leftrightarrow Y_1 \wedge \neg Y_2$$

X_3 corresponding “10”

$$X_3 \leftrightarrow \neg Y_1 \wedge Y_2$$

X_4 corresponding “11”

$$X_4 \leftrightarrow Y_1 \wedge Y_2$$

- AMO Encoding:

$$(X_1 \rightarrow \neg Y_1 \wedge \neg Y_2) \wedge (X_2 \rightarrow Y_1 \wedge \neg Y_2) \wedge (X_3 \rightarrow \neg Y_1 \wedge Y_2) \wedge (X_4 \rightarrow Y_1 \wedge Y_2)$$

- ALO Encoding:

$$(i): (\neg Y_1 \wedge \neg Y_2 \rightarrow X_1) \wedge (Y_1 \wedge \neg Y_2 \rightarrow X_2) \wedge (\neg Y_1 \wedge Y_2 \rightarrow X_3) \wedge (Y_1 \wedge Y_2 \rightarrow X_4)$$

$$(ii): (X_1 \vee X_2 \vee X_3 \vee X_4)$$

Binary encoding (2)

- For $n = 3$ (X_1, X_2, X_3)

X_1 corresponding “00”

$$X_1 \leftrightarrow \neg Y_1 \wedge \neg Y_2$$

X_2 corresponding “01”

$$X_2 \leftrightarrow Y_1 \wedge \neg Y_2$$

X_3 corresponding “10”

$$X_3 \leftrightarrow \neg Y_1 \wedge Y_2$$

- AMO Encoding:

$$(X_1 \rightarrow \neg Y_1 \wedge \neg Y_2) \wedge (X_2 \rightarrow Y_1 \wedge \neg Y_2) \wedge (X_3 \rightarrow \neg Y_1 \wedge Y_2)$$

- ALO Encoding: ($X_1 \vee X_2 \vee X_3$)

Binary encoding (3)

- Optimization (for $n = 3$)

X_1 corresponding “00” or “01” $X_1 \leftrightarrow (\neg Y_1 \wedge \neg Y_2) \vee (Y_1 \wedge \neg Y_2) \equiv \neg Y_2$

X_2 corresponding “10” $X_2 \leftrightarrow \neg Y_1 \wedge Y_2$

X_3 corresponding “11” $X_3 \leftrightarrow Y_1 \wedge Y_2$

- AMO Encoding:

$$(X_1 \rightarrow \neg Y_2) \wedge (X_2 \rightarrow \neg Y_1 \wedge Y_2) \wedge (X_3 \rightarrow Y_1 \wedge Y_2)$$

- ALO Encoding: $(X_1 \vee X_2 \vee X_3)$ or $(\neg Y_2 \rightarrow X_1) \wedge ((\neg Y_1 \wedge Y_2) \rightarrow X_2) \wedge ((Y_1 \wedge Y_2) \rightarrow X_3)$

Binary encoding (4)

- For $n = 5$ (X_1, X_2, X_3, X_4, X_5)

X_1 corresponding “000”

$$X_1 \leftrightarrow \neg Y_1 \wedge \neg Y_2 \wedge \neg Y_3$$

X_2 corresponding “001”

$$X_2 \leftrightarrow Y_1 \wedge \neg Y_2 \wedge \neg Y_3$$

X_3 corresponding “010”

$$X_3 \leftrightarrow \neg Y_1 \wedge Y_2 \wedge \neg Y_3$$

X_4 corresponding “100”

$$X_4 \leftrightarrow \neg Y_1 \wedge \neg Y_2 \wedge Y_3$$

X_5 corresponding “011”

$$X_5 \leftrightarrow Y_1 \wedge Y_2 \wedge \neg Y_3$$

- AMO Encoding:

$$(X_1 \rightarrow \neg Y_1 \wedge \neg Y_2 \wedge \neg Y_3) \wedge (X_2 \rightarrow Y_1 \wedge \neg Y_2 \wedge \neg Y_3) \wedge (X_3 \rightarrow \neg Y_1 \wedge Y_2 \wedge \neg Y_3) \wedge (X_4 \rightarrow \neg Y_1 \wedge \neg Y_2 \wedge Y_3) \wedge (X_5 \rightarrow Y_1 \wedge Y_2 \wedge \neg Y_3)$$

- ALO Encoding: ($X_1 \vee X_2 \vee X_3 \vee X_4 \vee X_5$)

→ X_1 corresponding “000” or “001” or “010” or “100”?

Sequential encounter encoding

- Idea:
 - Variables: $X_1, X_2, \dots, X_n$ and new $(n-1)$ variables $S_1, S_2, \dots, S_{n-1}$
 - $S_i = \text{True}$ iff existing a Truth variable in $(X_1, X_2, \dots, X_i)$
 - If $S_i = \text{True}$ then X_{i+1} false
 - AMO Clauses: $X_1 + X_2 + \dots + X_n \leq 1$
 - $X_1 \rightarrow S_1$
 - $(X_i \vee S_{i-1}) \rightarrow S_i \wedge (S_{i-1} \rightarrow \neg X_i) \ (1 < i < n)$
 - $\wedge (S_{n-1} \rightarrow \neg X_n)$
 - Requires: $(n-1)$ new variables, $(3n-4)$ clauses

Sequential encounter encoding (cont)

- For $n = 4$ (X_1, X_2, X_3, X_4)
- New variables S_1, S_2, S_3
- AMO Encoding:
 - $X_1 \rightarrow S_1 \wedge$
 - $(X_2 \vee S_1) \rightarrow S_2 \wedge (S_1 \rightarrow \neg X_2) \wedge$
 - $(X_3 \vee S_2) \rightarrow S_3 \wedge (S_2 \rightarrow \neg X_3) \wedge$
 - $(S_3 \rightarrow \neg X_4)$
- ALO Encoding: $(X_1 \vee X_2 \vee X_3 \vee X_4)$

Commander encoding

- Idea:
 - Partitioning set of variables $X_1, X_2, \dots, X_n$ into some groups of some fixed size and encoding AMO over each group
 - $X_1, X_2, \dots, X_k \in G_1$
 - $X_{k+1}, X_{k+2}, \dots, X_{2k} \in G_2; \dots$
 - $X_j, \dots, X_n \in G_m$
 - Introducing a commander variable for each group: $c_1, c_2, \dots, c_m$
 - AMO, ALO Clauses for $c_1, c_2, \dots, c_m$ (Binomial encoding)
 - If a commander variable c_i is true then exactly one variable true in G_i (AMO and ALO of G_i)
 - If a commander variable c_i is false then all variables false in G_i

Commander encoding (cont)

- For $n = 6$ ($X_1 X_2, \dots, X_6$)
 - $X_1, X_2, X_3 \in G_1$
 - $X_4, X_5, X_6 \in G_2$
 - $(c_1 \vee c_2) \wedge (c_1 \rightarrow \neg c_2) \wedge$
 - $c_1 \rightarrow (X_1 \vee X_2 \vee X_3) \wedge (X_1 \rightarrow \neg X_2) \wedge (X_1 \rightarrow \neg X_3) \wedge (X_2 \rightarrow \neg X_3) \wedge$
 - $c_2 \rightarrow (X_4 \vee X_5 \vee X_6) \wedge (X_4 \rightarrow \neg X_5) \wedge (X_4 \rightarrow \neg X_6) \wedge (X_5 \rightarrow \neg X_6) \wedge$
 - $(\neg c_1 \rightarrow \neg X_1) \wedge (\neg c_1 \rightarrow \neg X_2) \wedge (\neg c_1 \rightarrow \neg X_3) \wedge$
 - $(\neg c_2 \rightarrow \neg X_4) \wedge (\neg c_2 \rightarrow \neg X_5) \wedge (\neg c_2 \rightarrow \neg X_6)$
- For $n = 7$ ($X_1 X_2, \dots, X_7$)?
 - $X_1, X_2, X_3, X_4 \in G_1$
 - $X_5, X_6, X_7 \in G_2$

$$G_1 \Leftrightarrow c_1$$

$$G_2 \Leftrightarrow c_2$$

Product encoding

- Idea: Decompose AMO ($X_1, X_2, \dots, X_n$) into: AMO ($r_1, r_2, \dots, r_p$) and AMO ($c_1, c_2, \dots, c_q$)
 - A grid with n points ($p \times q$): $p = \lceil \sqrt{n} \rceil, q = \lceil n/p \rceil$
 - One point for each variable ($X_1, X_2, \dots, X_n$)
 - $X_1 \Leftrightarrow (r_1, c_1), X_2 \Leftrightarrow (r_1, c_2), \dots, X_n \Leftrightarrow (r_p, c_k)$ $k = n - (p-1) \cdot q$
 - Constraints at most one row and at most one column
- AMO ($X_1, X_2, \dots, X_n$):
 - AMO ($r_1, r_2, \dots, r_p$) and AMO ($c_1, c_2, \dots, c_q$)
 - $X_i \rightarrow r_u \wedge c_v$ where $X_i \Leftrightarrow (r_u, c_v)$
- ALO ($X_1, X_2, \dots, X_n$):
 - (i): ($X_1 \vee X_2 \vee X_3 \vee X_4$)
 - (ii): ALO ($r_1, r_2, \dots, r_p$) and ALO ($c_1, c_2, \dots, c_q$) and $r_u \wedge c_v \rightarrow X_i$ where $X_i \Leftrightarrow (r_u, c_v)$

Product encoding (1)

- For $n = 5$ (X_1, X_2, X_3, X_4, X_5)
 - $p = \lceil \sqrt{5} \rceil = 3, v = \lceil 5/3 \rceil = 2$
 - $(r_1, r_2, r_3) \times (c_1, c_2)$
- AMO (X_1, X_2, X_3, X_4, X_5):
 - $r_1 \rightarrow \neg r_2 \wedge r_1 \rightarrow \neg r_3 \wedge r_2 \rightarrow \neg r_3 \wedge$
 - $c_1 \rightarrow \neg c_2 \wedge$
 - $X_1 \rightarrow r_1 \wedge X_1 \rightarrow c_1 \wedge X_2 \rightarrow r_1 \wedge X_2 \rightarrow c_2 \wedge$
 - $X_3 \rightarrow r_2 \wedge X_3 \rightarrow c_1 \wedge X_4 \rightarrow r_2 \wedge X_4 \rightarrow c_2 \wedge$
 - $X_5 \rightarrow r_3 \wedge X_5 \rightarrow c_1$

Product encoding (2)

- For $n = 6$ ($X_1, X_2, X_3, X_4, X_5, X_6$)
 - $p = \lceil \sqrt{5} \rceil = 3, v = \lceil 5/3 \rceil = 2$
 - $(r_1, r_2, r_3) \times (c_1, c_2)$
- AMO ($X_1, X_2, X_3, X_4, X_5, X_6$):
 - $r_1 \rightarrow \neg r_2 \wedge r_1 \rightarrow \neg r_3 \wedge r_2 \rightarrow \neg r_3 \wedge$
 - $c_1 \rightarrow \neg c_2 \wedge$
 - $X_1 \rightarrow r_1 \wedge X_1 \rightarrow c_1 \wedge X_2 \rightarrow r_1 \wedge X_2 \rightarrow c_2 \wedge$
 - $X_3 \rightarrow r_2 \wedge X_3 \rightarrow c_1 \wedge X_4 \rightarrow r_2 \wedge X_4 \rightarrow c_2 \wedge$
 - $X_5 \rightarrow r_3 \wedge X_5 \rightarrow c_1 \wedge X_6 \rightarrow r_3 \wedge X_6 \rightarrow c_2$
- For $n = 7$ ($p \times q = 3 \times 3$)? $n = 8$ ($p \times q = 3 \times 3$)?

Questions?